

The Digestive System

From motility and digestion to the disorders that disrupt them — GERD and ulcers, IBD and obstruction, jaundice and cirrhosis, pancreatitis, and the pediatric GI emergencies.

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Module 13

Function · Disorders · Hepatobiliary

9 topics + cheat sheet

Structure & Function

SECTION 1

How food moves, gets broken down, and is absorbed — and the digestive juices that make it happen.

Food is moved by **peristalsis** — coordinated waves of smooth-muscle contraction — from mouth to anus. Sphincters control passage between segments.

Process	Mechanism
Swallowing / esophageal transit	Peristalsis (smooth-muscle waves) pushes the bolus down
Gastric emptying	A high-fat meal slows emptying (fat triggers CCK, which delays the stomach to allow digestion)
Defecation reflex	Rectal stretch triggers the urge; an impaired reflex (e.g., nerve damage) reduces the urge → retention/constipation
Autonomic control	Parasympathetic stimulates GI activity; anticholinergics block it → dry mouth, slowed motility

Key idea: the GI tract is a muscular tube run by the autonomic nervous system.
Parasympathetic = "rest and digest" (more secretion and motility).
Anticholinergic drugs block parasympathetic signals → dry mouth, slowed gut, constipation — because acetylcholine drives salivation and peristalsis.

PRACTICE

Q1. Which mechanism allows food to move through the esophagus?

- A. Peristalsis (waves of smooth-muscle contraction)
- B. Gravity alone
- C. Segmentation in the stomach
- D. Bile secretion

A is correct. Coordinated peristaltic waves propel the bolus down the esophagus — which is why swallowing works even lying down.

PRACTICE

Q2. A patient on an anticholinergic medication reports dry mouth. Which mechanism explains it?

- A. Blocked parasympathetic (acetylcholine) signaling reduces salivation
- B. Increased parasympathetic activity
- C. Excess salivary secretion
- D. Increased peristalsis

A is correct. Anticholinergics block acetylcholine, the parasympathetic transmitter that drives salivation and gut motility — so the mouth goes dry and the gut slows.

PRACTICE

Q3. A patient has delayed gastric emptying after a high-fat meal. What contributes to the delay?

- A. Fat in the duodenum triggers signals (CCK) that slow gastric emptying
- B. Fat speeds gastric emptying
- C. Fat increases peristalsis in the stomach
- D. Fat is absorbed in the stomach

A is correct. Fat entering the duodenum triggers CCK, which slows gastric emptying so there's time to digest and absorb the fat.

2 Secretion & Digestion



Digestion runs on secretions — acid, enzymes, bicarbonate, and bile — each from a specific source.

Secretion (source)	Job
Gastric acid + intrinsic factor (parietal cells)	Acid digests/sterilizes; intrinsic factor is needed to absorb vitamin B12
Pancreatic enzymes	Digest carbs, protein, and fat in the duodenum
Pancreatic bicarbonate	Neutralizes stomach acid entering the duodenum (protects the lining; lets enzymes work)
Bile (liver; stored in gallbladder)	Emulsifies fat — breaks large fat globules into droplets so enzymes can digest them
Gut microbiome	Supports digestion, vitamin production, and immune balance; low diversity impairs these

Key idea — parietal cells: they make both **acid** and **intrinsic factor**. Damage them and you get **low acid** and **B12 malabsorption** (intrinsic factor is required to absorb B12 in the ileum) → pernicious anemia.

Pearl — bicarbonate & bile: pancreatic **bicarbonate** neutralizes acid in the duodenum — lose it and the duodenum becomes too acidic (enzymes fail, lining is injured). **Bile emulsifies fat** — without it, fat isn't digested/absorbed → fatty stools (steatorrhea).

PRACTICE

Q1. A patient has damage to gastric parietal cells. Which change is expected?

- A. Decreased acid and impaired vitamin B12 absorption (loss of intrinsic factor)
- B. Increased acid secretion
- C. Increased intrinsic factor

D. Improved B12 absorption

A is correct. Parietal cells make acid *and* intrinsic factor; losing them lowers acid and impairs B12 absorption → pernicious anemia.

PRACTICE

Q2. What is the role of bile in fat digestion?

- A. It emulsifies fat into smaller droplets for enzyme action
- B. It chemically digests fat into fatty acids
- C. It neutralizes stomach acid
- D. It absorbs fat directly

A is correct. Bile emulsifies (physically breaks up) fat into droplets, increasing surface area so pancreatic lipase can digest it. It doesn't chemically digest fat itself.

PRACTICE

Q3. A patient has decreased pancreatic bicarbonate secretion. Which effect occurs in the duodenum?

- A. The duodenum becomes too acidic, impairing enzymes and injuring the lining
- B. The duodenum becomes more alkaline
- C. Fat absorption improves
- D. Acid is fully neutralized

A is correct. Without bicarbonate to neutralize gastric acid, the duodenum stays acidic — digestive enzymes work poorly and the mucosa is at risk.

The small intestine absorbs nutrients across its huge surface of **villi**. Damage the villi or the digestive juices, and nutrients pass through unabsorbed.

Disorder	Mechanism	Result
Celiac disease	Immune reaction to gluten flattens/shortens the villi	↓ absorptive surface → chronic diarrhea, weight loss, deficiencies
Cystic fibrosis	Thick secretions block pancreatic ducts → enzymes can't reach the gut	Fat malabsorption → steatorrhea, poor growth
Pancreatic insufficiency	Too few pancreatic enzymes (incl. lipase)	Steatorrhea (fatty, floating, foul stools)

Key idea: absorption needs (1) **working villi** (surface area) and (2) **digestive juices** (enzymes + bile). **Celiac** destroys the villi (a *surface* problem). **Cystic fibrosis / pancreatic insufficiency** starve the gut of enzymes (a *digestion* problem) → especially **fat malabsorption (steatorrhea)**.

PRACTICE

Q1. A patient with celiac disease has chronic diarrhea and weight loss. Which physiologic outcome from shortened intestinal villi explains it?

- A. Reduced absorptive surface area causing malabsorption
- B. Increased surface area for absorption
- C. Increased enzyme secretion
- D. Faster nutrient uptake

A is correct. Flattened villi sharply reduce absorptive surface area, so nutrients (and water) aren't absorbed → diarrhea and weight loss.

PRACTICE

Q2. Which alteration explains fat malabsorption in cystic fibrosis?

- A. Blocked pancreatic ducts prevent digestive enzymes from reaching the intestine
- B. Excess pancreatic enzyme secretion
- C. Increased bile production
- D. Enhanced villous absorption

A is correct. Thick secretions plug the pancreatic ducts, so enzymes (including lipase) can't reach the duodenum → fat isn't digested/absorbed → steatorrhea.

Gastrointestinal Disorders

SECTION 2

Upper-GI acid and bleeding problems, the inflammatory bowel diseases, and mechanical obstruction.

The upper GI story is mostly about **acid where it shouldn't be** and **bleeding**.

Disorder	Mechanism
GERD	Weak/relaxed lower esophageal sphincter (LES) lets acid reflux up into the esophagus. Risk: obesity, hiatal hernia, fatty/large meals, lying down after eating
Peptic ulcer disease (PUD)	Mucosal defense fails against acid — mainly H. pylori infection and NSAIDs (also smoking, alcohol, stress)
Esophageal varices	Portal hypertension swells esophageal veins → can rupture → massive bright-red hematemesis

Reading GI bleeding

- **Hematemesis** (vomiting blood): **bright red** = brisk/upper bleeding (e.g., varices); "coffee-ground" = older, acid-exposed blood.
- **Melena** (black, tarry stool): **upper GI bleed** — blood has been digested on its way through.
- **Hematochezia** (bright red rectal blood): usually a **lower GI** source.

Key idea: GERD = **LES failure** (acid refluxes up). PUD = **H. pylori + NSAIDs** breaking down mucosal defense. For bleeding, location is in the color: **black tarry stool (melena) = upper GI bleed**; bright-red rectal blood = lower GI.

Pearl: prolonged vomiting loses stomach acid → **metabolic alkalosis** plus fluid/electrolyte loss (a common exam complication of severe vomiting).

PRACTICE

Q1. Which change allows gastric contents to reflux into the esophagus in GERD?

- A. A weakened/relaxed lower esophageal sphincter
- B. A tightened LES
- C. Increased esophageal peristalsis

D. Decreased gastric acid

A is correct. When the LES is weak or relaxes inappropriately, acidic stomach contents flow back up — the core of GERD.

PRACTICE

Q2 (SATA). Which factors contribute to peptic ulcer disease?

- A. H. pylori infection
- B. NSAID use
- C. Smoking
- D. Increased mucosal protection
- E. Excess acid/stress

A, B, C, and E are correct. H. pylori, NSAIDs, smoking, and acid/stress all weaken or overwhelm mucosal defense. (D) is wrong — *increased* protection would prevent ulcers.

PRACTICE

Q3. A patient reports black, tarry stools (melena). What does this indicate?

- A. An upper GI bleed (blood digested in transit)
- B. A lower GI/rectal bleed
- C. Normal stool
- D. Biliary obstruction

A is correct. Melena is black and tarry because blood from an upper GI source is partly digested as it moves through the tract.

Inflammation can be localized (appendicitis, diverticulitis) or chronic (IBD). The dreaded shared complication is **peritonitis**.

Crohn disease vs ulcerative colitis (IBD)

	Crohn disease	Ulcerative colitis (UC)
Location	Anywhere mouth → anus; skip lesions	Colon/rectum only; continuous
Depth	Transmural (full wall) → fistulas, strictures	Mucosa/submucosa only
Classic stool	Diarrhea (less bloody)	Bloody diarrhea with urgency
Cancer risk	Increased	Higher colorectal cancer risk → surveillance colonoscopy

Localized inflammation & diarrhea

- **Appendicitis:** RLQ pain, fever, nausea. If it ruptures → **peritonitis:** diffuse pain, **rigid abdomen**, rebound tenderness — an emergency.
- **Diverticulitis:** LLQ pain, fever, nausea (inflamed colonic outpouchings).
- **Secretory diarrhea** keeps going **even when fasting** (the gut actively secretes fluid); osmotic diarrhea stops with fasting.
- **Top complication of persistent diarrhea: dehydration and electrolyte imbalance.**

Key idea — Crohn vs UC: Crohn = skip lesions, anywhere, transmural (deep → fistulas). UC = continuous, colon-only, mucosal, bloody diarrhea with higher colorectal-cancer risk. The single best distinguisher: Crohn can affect *any* part of the GI tract with skip lesions; UC is limited to the colon and is continuous.

PRACTICE

Q1. Which statement distinguishes Crohn disease from ulcerative colitis?

- A. Crohn can affect any part of the GI tract with skip lesions; UC is continuous and colon-only
- B. UC affects the entire GI tract; Crohn affects only the rectum
- C. Both are limited to the colon
- D. Crohn is mucosal only

A is correct. Crohn = skip lesions anywhere, transmural; UC = continuous, colon/rectum only, mucosal, with bloody diarrhea.

PRACTICE

Q2. A patient with appendicitis develops severe diffuse abdominal pain, fever, and a rigid abdomen. What happened?

- A. The appendix ruptured, causing peritonitis
- B. The appendix healed
- C. A bowel obstruction resolved
- D. Gallbladder inflammation

A is correct. Rupture spills contents into the peritoneum → peritonitis: diffuse pain, rigid (guarding) abdomen, rebound tenderness, fever — a surgical emergency.

PRACTICE

Q3. Which complication poses the highest risk in a patient with persistent large-volume diarrhea?

- A. Dehydration and electrolyte imbalance
- B. Weight gain
- C. Hypertension
- D. Polycythemia

A is correct. Large fluid and electrolyte losses cause dehydration and imbalances (e.g., hypokalemia) — the main danger of persistent diarrhea.



When the lumen is blocked, contents back up. The classic picture: **distension, cramping pain, vomiting, and no flatus/stool**.

Cause	Mechanism
Mechanical obstruction	Adhesions, hernia, tumor, etc. block the lumen → distension, cramping, vomiting, no flatus ; bowel sounds first high-pitched/hyperactive , then absent
Intussusception	One segment telescopes into another → obstruction + compromised blood flow → currant-jelly stools (child)
Hirschsprung disease	Absent ganglion cells in a distal colon segment → no peristalsis there → functional obstruction → constipation (newborn)

Key idea: obstruction blocks forward flow → gas and fluid accumulate → **distension, colicky pain, vomiting, and obstipation (no flatus/stool)**. Early high-pitched "tinkling" bowel sounds reflect the gut straining against the block. A feared complication is **bowel ischemia/perforation** → **peritonitis**.

Pearl — pediatric obstructions: **intussusception** = telescoping → "currant-jelly" (blood + mucus) stools. **Hirschsprung** = *missing* nerve (ganglion) cells → that segment can't relax/propel → constipation from birth.

PRACTICE

Q1. A patient has abdominal distension, vomiting, cramping pain, and no flatus. What process is occurring?

- A. Bowel obstruction
- B. Diarrhea
- C. GERD
- D. Normal digestion

A is correct. Distension + cramping + vomiting + absent flatus is the classic bowel-obstruction picture (Marco's case). Contents can't move forward, so they back up.

PRACTICE

Q2. A child has intermittent severe abdominal pain, vomiting, and currant-jelly stools. Which process is most likely?

- A. Intussusception (one bowel segment telescopes into another)
- B. GERD
- C. Appendicitis
- D. Celiac disease

A is correct. Telescoping bowel obstructs flow and compromises blood supply, producing the classic currant-jelly (blood + mucus) stools.

PRACTICE

Q3. Which change causes constipation in Hirschsprung disease?

- A. Absent ganglion (nerve) cells in a colon segment prevent peristalsis
- B. Excess ganglion cells speed peristalsis
- C. Increased mucus secretion
- D. Shortened villi

A is correct. The aganglionic segment can't relax or propel stool, creating a functional obstruction → constipation from birth.

Hepatobiliary & Pediatric GI

SECTION 3

The liver, gallbladder, and pancreas — plus the pediatric GI conditions you must recognize.

The liver processes **bilirubin**, makes proteins, and filters portal blood. Its failure shows up as jaundice, fluid shifts, and portal hypertension.

Jaundice — where bilirubin gets stuck

Type	Mechanism
Pre-hepatic (hemolytic)	Too much RBC breakdown → excess unconjugated bilirubin overwhelms the liver
Hepatic	Damaged liver cells can't process/conjugate bilirubin
Post-hepatic (cholestatic)	Blocked bile flow → conjugated bilirubin backs up → dark urine, pale stools, and pruritus (bile salts in skin)

Cirrhosis & portal hypertension

- **Cirrhosis:** chronic injury (alcohol, hepatitis, fatty liver) → repeated inflammation → **fibrosis/scarring** replaces functional liver → progressive failure.
- **Portal hypertension:** scarring (or portal vein obstruction) blocks blood flow through the liver → back-pressure in the portal system → **esophageal varices, splenomegaly, ascites**.
- **Ascites:** fluid in the abdomen from **high portal pressure + low albumin** (failing liver makes less albumin → low oncotic pressure).

Key idea: jaundice = bilirubin backing up somewhere along its path (overproduction, liver can't process it, or bile is blocked). **Cholestasis** (blocked bile) gives the triad **yellow skin + dark urine + pale stools**, plus **itching** from bile salts. Cirrhosis → portal hypertension → varices + ascites; low albumin worsens the ascites.

PRACTICE

Q1. A patient reports yellowing skin, dark urine, and pale stools. Which mechanism explains it?

A. Blocked bile flow backs up conjugated bilirubin (cholestasis)

- B. Increased bile excretion
- C. Excess RBC production
- D. Increased albumin

A is correct. Obstructed bile flow makes conjugated bilirubin back up into blood (jaundice, dark urine) while it can't reach the gut (pale stools).

PRACTICE

Q2. Which mechanism explains hyperbilirubinemia in hemolysis?

- A. Excessive RBC breakdown produces more bilirubin than the liver can process
- B. Blocked bile ducts
- C. Reduced RBC production
- D. Increased liver conjugation

A is correct. Rapid RBC destruction floods the system with unconjugated bilirubin, overwhelming the liver's processing capacity.

PRACTICE

Q3. Which mechanism explains ascites in liver failure?

- A. Portal hypertension plus low albumin (low oncotic pressure) shift fluid into the abdomen
- B. Increased albumin production
- C. Low portal pressure
- D. Increased bile flow

A is correct. High portal pressure pushes fluid out, and a failing liver makes less albumin (low oncotic pull to keep fluid in vessels) → ascites.



The gallbladder stores bile; the pancreas makes enzymes and bicarbonate. Both produce classic pain syndromes.

Disorder	Mechanism & clues
Cholelithiasis / cholecystitis	Gallstones block bile flow / inflame the gallbladder → RUQ pain after fatty meals , fever, nausea (fat triggers gallbladder contraction)
Pancreatitis	Pancreatic enzymes activate inside the pancreas and autodigest it → severe epigastric pain (often radiating to the back), nausea
Pancreatic insufficiency	Too few enzymes (incl. lipase) → fat malabsorption → steatorrhea

Key idea — pancreatitis is autodigestion: enzymes meant for the gut activate prematurely **within the pancreas** and digest the organ itself → intense inflammation and pain. Common causes: **gallstones and alcohol**. If the gland is chronically damaged, it can't secrete enzymes → **steatorrhea** (Topic 3 link).

Pearl — "after a fatty meal": fat triggers gallbladder contraction (via CCK), so **RUQ pain after fatty meals** points to gallstones/cholecystitis. **Pruritus in cholestasis** comes from **bile salts** deposited in the skin.

PRACTICE

Q1. Which mechanism explains the abdominal pain of pancreatitis?

- A. Premature enzyme activation autodigesting the pancreas and causing inflammation
- B. Gallbladder contraction
- C. Decreased pancreatic enzymes
- D. Esophageal reflux

A is correct. Enzymes activate inside the pancreas and digest its own tissue
→ intense inflammation and pain.

PRACTICE

Q2. A patient develops RUQ pain after meals with fever and nausea. What is the likely problem?

- A. Cholecystitis/gallstones (fatty meals trigger gallbladder contraction)
- B. Appendicitis
- C. Diverticulitis
- D. GERD

A is correct. RUQ pain after fatty meals with fever/nausea is the classic gallbladder pattern — fat triggers contraction against an obstructing stone.

PRACTICE

Q3. Which mechanism explains steatorrhea in pancreatic insufficiency?

- A. Lack of pancreatic enzymes (lipase) leaves fat undigested and unabsorbed
- B. Excess bile
- C. Increased villous absorption
- D. Too much lipase

A is correct. Without lipase, fat isn't digested or absorbed, so it passes into the stool → fatty, floating, foul-smelling steatorrhea.

Several GI conditions are classic in newborns and children — recognize the pattern and the mechanism.

Condition	Mechanism & hallmark
Pyloric stenosis	Thickened pyloric muscle obstructs gastric outlet → forceful, projectile, NON-bilious vomiting in an infant (block is <i>before</i> the bile duct entry)
Intussusception	Bowel telescopes → obstruction + ischemia → currant-jelly stools , intermittent severe pain
Hirschsprung disease	Absent ganglion cells → no peristalsis in that segment → constipation/obstruction from birth
Necrotizing enterocolitis (NEC)	Premature infant; bowel inflammation/ischemia → abdominal distension, feeding intolerance, bloody stools — emergency
Cleft palate	Opening in the palate → difficult feeding, poor weight gain (can't generate suction)

Key idea — "non-bilious" vs "bilious": in **pyloric stenosis** the obstruction is at the stomach outlet, *above* where bile enters → vomit is **non-bilious** (and projectile). Obstructions *below* the bile duct produce **bilious (green)** vomiting. Where the blockage sits tells you the vomit's color.

Pearl — NEC: think **premature infant + distension + feeding intolerance + bloody stools**. Immature gut and immunity make preemies vulnerable to this dangerous bowel inflammation.

PRACTICE

Q1. Which change explains forceful, non-bilious vomiting in an infant?

A. Pyloric muscle thickening obstructs the gastric outlet (above the bile duct)

- B. A lower bowel obstruction
- C. Bile reflux into the stomach
- D. Esophageal sphincter relaxation

A is correct. Pyloric stenosis blocks the stomach outlet before bile enters the gut, so vomiting is forceful (projectile) and non-bilious.

PRACTICE

Q2. A premature infant has abdominal distension, feeding intolerance, and bloody stools. Which condition is most consistent?

- A. Necrotizing enterocolitis (NEC)
- B. GERD
- C. Cleft palate
- D. Pyloric stenosis

A is correct. Prematurity + distension + feeding intolerance + bloody stools is the classic NEC picture — a GI emergency.

PRACTICE

Q3. An infant with cleft palate has difficult feeding and poor weight gain. Why?

- A. The palate opening prevents effective suction during feeding
- B. The stomach can't empty
- C. Bile flow is blocked
- D. The villi are flattened

A is correct. A cleft palate prevents the infant from creating the suction/seal needed to feed effectively → poor intake and weight gain.

Digestive Cheat Sheet

The highest-yield patterns — review the night before.

Function

- **Peristalsis** moves food; **anticholinergics** block parasympathetic → dry mouth, slow gut; **high-fat meal** slows gastric emptying (CCK).
- **Parietal cells** = acid + **intrinsic factor** (needed for B12). **Pancreatic bicarbonate** neutralizes duodenal acid. **Bile emulsifies fat**.
- **Celiac** = flattened villi (surface loss). **CF / pancreatic insufficiency** = no enzymes → **steatorrhea**.

GI disorders

- **GERD** = weak LES → acid reflux. **PUD** = H. pylori + NSAIDs.
- **Melena** (black tarry) = **upper** GI bleed; bright-red rectal = lower. Bright-red hematemesis = varices.
- **Crohn** = skip lesions, anywhere, transmural. **UC** = continuous, colon-only, mucosal, **bloody diarrhea**, ↑ colorectal cancer.
- **Appendicitis** (RLQ) rupture → **peritonitis** (rigid abdomen, rebound). **Diverticulitis** = LLQ.
- **Obstruction** = distension + cramping + vomiting + no flatus; high-pitched then absent bowel sounds.
- **Persistent diarrhea** → dehydration/electrolyte loss. **Secretory diarrhea** continues when fasting.

Hepatobiliary & pediatric

- **Jaundice**: overproduction (hemolysis), liver can't process, or blocked bile (cholestasis → dark urine, pale stools, **itching**).
- **Cirrhosis** → fibrosis → **portal hypertension** → varices + **ascites** (also low albumin).
- **Pancreatitis** = autodigestion (gallstones, alcohol). **Cholecystitis** = RUQ pain after fatty meals.

- **Pyloric stenosis** = projectile **non-bilious** vomiting. **Intussusception** = currant-jelly stools. **Hirschsprung** = no ganglion cells → constipation. **NEC** = premature + distension + bloody stools.

□ TERMS

Glossary

Quick definitions of the Week 13 vocabulary.

Peristalsis

Waves of smooth-muscle contraction that propel GI contents.

Intrinsic factor

Parietal-cell protein required to absorb vitamin B12.

Bile

Liver secretion that emulsifies fat for digestion; stored in the gallbladder.

Steatorrhea

Fatty, foul, floating stool from fat malabsorption (pancreatic insufficiency, CF).

GERD

Acid reflux from a weak lower esophageal sphincter.

Peptic ulcer disease

Mucosal ulceration, mainly from *H. pylori* and NSAIDs.

Melena / Hematemesis

Black tarry stool (upper GI bleed) / vomiting blood.

Crohn vs UC

Crohn: skip lesions, anywhere, transmural. UC: continuous, colon-only, mucosal, bloody diarrhea.

Peritonitis

Inflammation of the peritoneum (e.g., ruptured appendix) — rigid abdomen, rebound tenderness.

Bowel obstruction

Blocked lumen → distension, cramping, vomiting, no flatus.

Intussusception

Bowel telescoping into itself → obstruction and currant-jelly stools (children).

Hirschsprung disease

Absent colonic ganglion cells → no peristalsis → constipation from birth.

Jaundice

Yellowing from elevated bilirubin (overproduction, hepatic, or obstructive).

Cirrhosis

Scarring/fibrosis from chronic liver injury → liver failure and portal hypertension.

Portal hypertension / Ascites

High pressure in the portal system → varices, splenomegaly, abdominal fluid (ascites).

Pancreatitis

Autodigestion of the pancreas by its own prematurely activated enzymes.

NEC

Necrotizing enterocolitis — bowel inflammation/ischemia in premature infants.